

Calibrating Expected Goals: An Out-of-Sample Test of xG-Based Match Probabilities Against Pinnacle Closing Lines

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Abstract

We test whether team ratings built from expected goals (xG) produce better probabilistic forecasts of football match outcomes than a classical goals-based model, using Pinnacle closing odds as the efficiency benchmark. Our sample covers 8,982 matches from Europe's top five leagues across the 2020/21 to 2024/25 seasons. We join historical Pinnacle closing 1X2 prices with team-level expected goals from the dribble360 football data API and generate strictly out-of-sample forecasts from two walk-forward models: a time-decayed Dixon-Coles goal model and a Poisson model driven by time-weighted xG ratings. Both models are well calibrated, with expected calibration errors of 0.014 and 0.035. The goals-based Dixon-Coles model is marginally sharper than the xG variant on log loss (1.001 vs 1.017) and Brier score (0.597 vs 0.609), while Pinnacle's closing implied probabilities remain the sharpest of all (log loss 0.969, Brier 0.576). A flat-stake backtest wagering on model-identified value at closing prices loses 4 to 7 percent ROI with significant negative t-stats, consistent with strong-form efficiency of the football closing line. We conclude that xG ratings yield well-calibrated probabilities but, in the simple time-weighted specification tested here, do not improve on a well-fit goals model and do not beat the market.

Keywords: expected goals, football forecasting, Dixon-Coles, Poisson model, calibration, betting market efficiency, Pinnacle.

1. Introduction

Probabilistic forecasting of association football outcomes has a long quantitative tradition. Maher (1982) introduced the independent Poisson goal model, and Dixon and Coles (1997) extended it with a time-decay weighting and a correction for dependence in low-scoring matches, a specification that remains the workhorse of the field. Rue and Salvesen (2000) developed a Bayesian dynamic alternative. In parallel, expected goals (xG), which values each shot by its estimated scoring probability, has become the standard measure of chance quality in professional analytics, on the argument that shot volumes and locations carry more signal about team strength than goals alone.

Whether xG-based team ratings actually forecast better than goals-based ratings is less settled than practitioners sometimes assume. Goals are noisy, but xG models introduce their own estimation error, and the mapping from chance creation to goals scored is itself stochastic. The natural benchmark for any forecasting model is the betting market, and specifically the Pinnacle closing line, which the sports forecasting literature widely treats as the most efficient publicly observable probability estimate.

This paper makes three contributions. First, we assemble a large out-of-sample test set: 8,982 matches from the English Premier League, Spanish La Liga, Italian Serie A, German Bundesliga, and French Ligue 1 over five seasons (2020/21 to 2024/25), each with Pinnacle closing 1X2 odds and team-level xG joined from an independent data provider. Second, we compare two walk-forward forecasting models head to head: a time-decayed Dixon-Coles model fit on goals, and a Poisson model driven by time-weighted xG attack and defense ratings. Every forecast uses only information available before the match date. Third, we evaluate both models on proper scoring rules, calibration, and a wagering backtest against the closing line.

Our headline findings are sober. Both models are well calibrated. The goals-based Dixon-Coles model is slightly sharper than the xG variant, and neither approaches the sharpness of Pinnacle's closing prices. Wagering on model-identified value at closing odds loses money at a statistically significant rate. xG ratings produce honest probabilities, but they do not, in the specification tested here, beat a well-fit goals model or the market.

2. Data

Match results and historical odds come from football-data.co.uk, which publishes free historical CSV files covering more than 25 leagues and 20 seasons. We use the 2020/21 through 2024/25 seasons for the five leagues above: 8,982 matches in total, each with full-time scoreline and Pinnacle opening and closing 1X2 odds, alongside prices from other bookmakers.

Team expected goals come from the dribble360 football data API, which serves Opta-sourced match aggregates. For each team and match we pull expected goals scored and expected goals conceded. Because the two sources use different team naming conventions, we join on match date plus team name through a hand-verified 132-name alias map. As a validation check, we compare the dribble360 scoreline against the odds-source scoreline wherever both record goals: agreement is 99.98 percent across 4,594 matched games, confirming the join. Both teams' xG is available for 8,454 of the 8,982 matches (94.1 percent); the remainder lack team-level stat rows in the API archive.

3. Methods

3.1 Walk-forward protocol

For each league, matches are ordered by date. A match is forecast only if both teams have at least six prior league matches in the training history, which removes the unreliable early-season period. The first forecast dates fall in autumn 2020. All model parameters for a forecast are

estimated from matches played strictly before the forecast date. This yields 8,483 out-of-sample forecasts for the goals model and 8,005 for the xG model (the difference reflects xG availability).

3.2 Model 1: time-decayed Dixon-Coles

Following Dixon and Coles (1997), home goals and away goals are modelled as conditionally independent Poisson variables with means

$$\lambda = \exp(\text{home} + \text{attack}_h + \text{defense}_a), \mu = \exp(\text{attack}_a + \text{defense}_h),$$

with a multiplicative adjustment governed by a dependence parameter ρ for the four lowest scorelines. Match contributions to the likelihood are weighted by $\exp(-\xi * \text{days})$, with $\xi = 0.002$, so recent matches count more. Parameters are estimated by maximum likelihood under the constraint that mean attack strength is zero, refit monthly within each league. Outcome probabilities come from the Poisson scoreline matrix truncated at ten goals per team.

3.3 Model 2: xG Poisson

Each team's attack rating is its time-weighted average xG scored and its defense rating its time-weighted average xG conceded, using the same decay $\xi = 0.002$ over prior matches. The home advantage is estimated as the mean home-minus-away xG difference over the trailing 400 league matches. Expected goals for the fixture are $(\text{attack}_h + \text{defense}_a)/2$ plus or minus half the home edge, and outcome probabilities again come from the Poisson scoreline matrix. This is deliberately a simple specification: it asks whether plain xG averages, without further modelling, improve on goals.

3.4 Evaluation

We report log loss and Brier score for each model against the same metrics computed from Pinnacle closing implied probabilities (overround removed by normalization). Calibration is assessed with a pooled reliability diagram over all match-outcome pairs and summarized by the expected calibration error (ECE) across decile bins. Finally, we run a flat-stake backtest: bet one unit on any outcome where the model probability exceeds the Pinnacle implied probability by a threshold edge (0, 2, and 5 percent), settling at Pinnacle closing odds, and report ROI with t-stats.

4. Results

4.1 Sharpness

Table 1 reports proper scores. Pinnacle's closing line is the sharpest forecaster by a clear margin, as expected. Between the two models, the goals-based Dixon-Coles model is marginally sharper than the xG Poisson on both log loss and Brier score. The gap is small but consistent: modeling goals directly with a properly fit likelihood beats modelling smoothed xG averages in this sample.

Table 1: Out-of-sample accuracy vs the Pinnacle closing line.

Model	n	Log loss	Brier	ECE	Pin log loss	Pin Brier
Dixon-Coles (goals)	8,483	1.001	0.597	0.014	0.969	0.576
xG Poisson	8,005	1.017	0.609	0.035	0.975	0.580

4.2 Calibration

Figure 1 shows the reliability diagram. Both models track the diagonal closely through the middle of the probability range. The Dixon-Coles model is nearly perfectly calibrated (ECE 0.014). The xG model is slightly overconfident at high probabilities and underconfident at low ones (ECE 0.035), but remains far better calibrated than an uncalibrated heuristic would be. In practical terms, both models' stated probabilities mean what they say.

4.3 Backtest against the closing line

Table 2 and Figure 2 show the wagering results. At every edge threshold, betting model-identified value at Pinnacle closing prices loses money. The Dixon-Coles model loses 4.3 to 5.5 percent ROI over roughly ten to twelve thousand bets; the xG model loses 6.6 to 6.9 percent. All t-stats are negative and significant. The equity curves decline steadily rather than in jumps, indicating a systematic shortfall against the closing line rather than a few bad beats. This is the standard signature of a market that has already incorporated the information in these models.

Table 2: Flat-stake backtest, 1 unit per bet, settled at Pinnacle closing odds. A bet is placed when model probability $\geq (1 + \text{edge}) * \text{Pinnacle implied probability}$.

Model	Edge	Bets	Strike	Profit (units)	ROI	t-stat
Dixon-Coles	0%	12,391	30.7%	-564.3	-4.6%	-2.89
Dixon-Coles	2%	11,284	30.2%	-487.6	-4.3%	-2.57
Dixon-Coles	5%	9,712	28.8%	-538.0	-5.5%	-3.03
xG Poisson	0%	12,001	22.9%	-795.5	-6.6%	-3.73
xG Poisson	2%	11,294	22.4%	-746.6	-6.6%	-3.56
xG Poisson	5%	10,327	21.5%	-711.8	-6.9%	-3.48

5. Discussion

Three points deserve emphasis. First, calibration and sharpness are different virtues, and both models have the first while lacking the second relative to the market. A well-calibrated model is still useful for risk management, simulation, and pricing derivatives of match outcomes even when it cannot beat the closing line.

Second, the xG model's failure to beat the goals model should be read narrowly. We tested simple time-weighted xG averages, not an xG-native likelihood or a model that uses shot-level detail. Richer xG specifications, including separate ratings for open play and set pieces or score-state adjustments, might do better. Our result is that xG does not help by default; it must earn its place through modelling.

Third, the backtest uses closing odds, the hardest benchmark in sports wagering. Losing to the close does not imply the models contain no information; it implies the market already contains it. An interesting extension would test the models against opening odds to measure closing line value, which requires timestamped odds movement data we do not have.

Limitations include the restriction to Europe's top five leagues, the use of a single xG provider's aggregates, and the simple form of the xG model noted above.

6. Conclusion

On 8,483 out-of-sample forecasts across five seasons, expected-goals team ratings produce well-calibrated match probabilities but do not improve on a classical time-decayed Dixon-Coles goal model, and neither model beats Pinnacle's closing line. The closing line remains the best available probability estimate, and the wagering results are consistent with strong-form market efficiency. For practitioners, the message is that xG is a sound input to calibrated forecasting systems, not a shortcut around the market.

References

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- Maher, M. J. (1982). Modelling association football scores. *Statistica Neerlandica*, 36(3), 109-118.
- Rue, H., and Salvesen, O. (2000). Prediction and retrospective analysis of soccer matches in a league. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 49(3), 399-418.

Data and code availability

Match results and historical odds: football-data.co.uk (free download). Team expected goals: dribble360 API (<https://dribble360.com>). Analysis code and derived results are available in the public repository accompanying this paper. Raw Opta-sourced event aggregates are not redistributed; only code and derived outputs are published, per the data license.

Figure 1

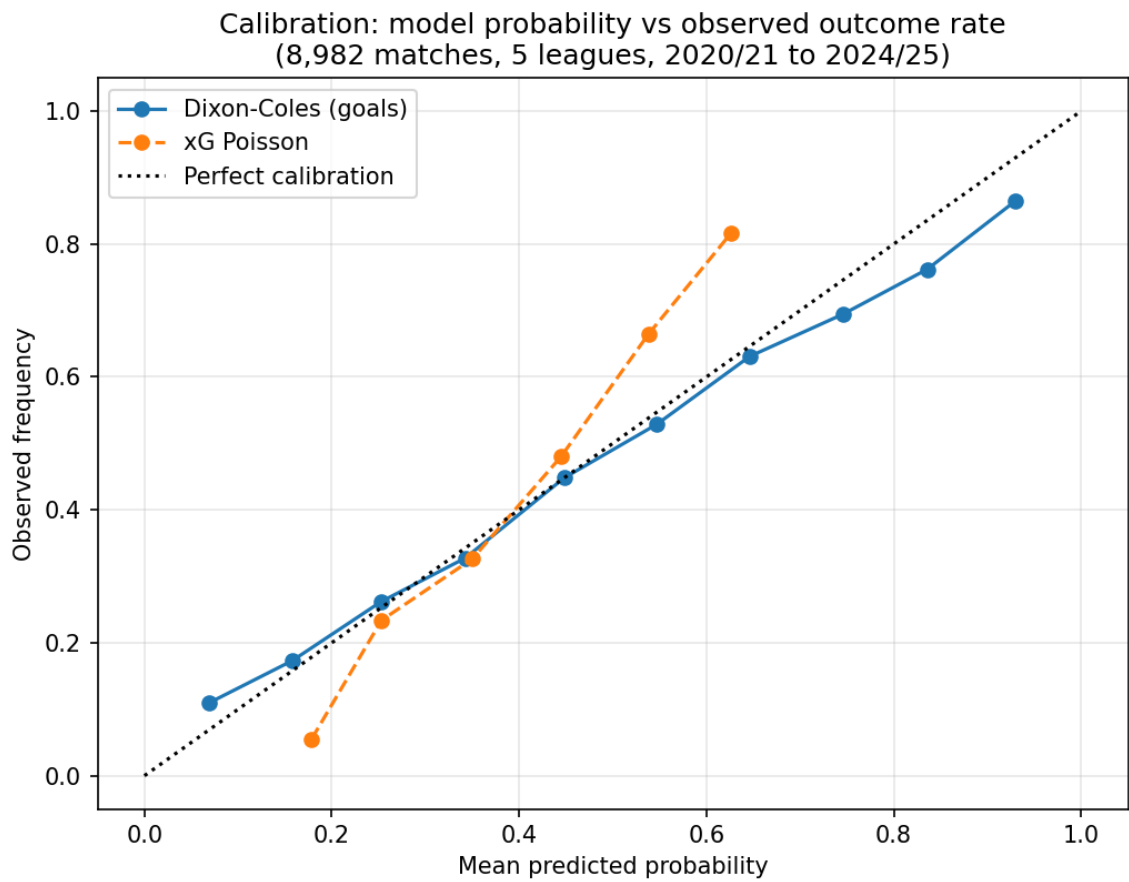


Figure 1: Reliability diagram. Mean predicted probability vs observed outcome frequency, pooled over all match-outcome pairs. The Dixon-Coles model (ECE 0.014) tracks the diagonal closely; the xG Poisson (ECE 0.035) is slightly overconfident at high probabilities.

Figure 2

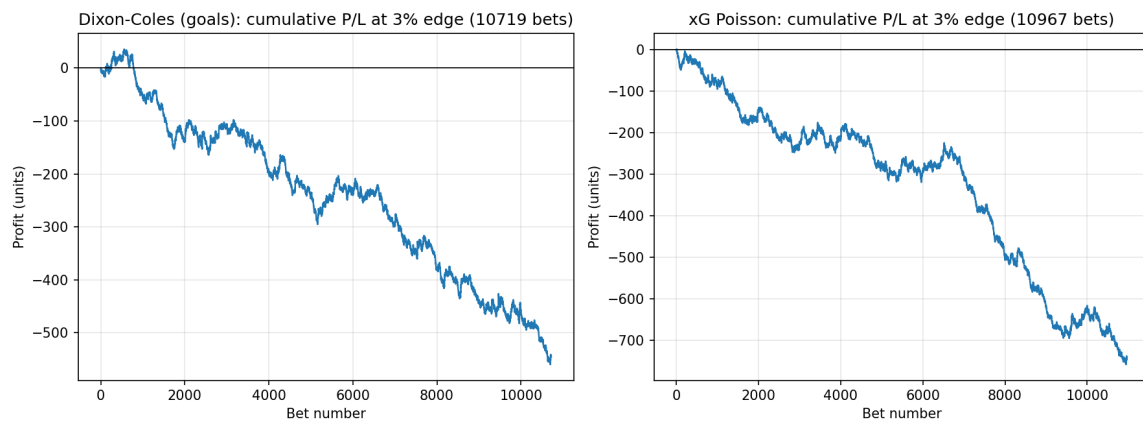


Figure 2: Cumulative profit/loss of the flat-stake backtest at a 3 percent edge threshold, settled at Pinnacle closing odds. Both equity curves decline steadily, consistent with strong-form efficiency of the closing line.